THE JOURNAL OF NERVOUS AND MENTAL DISEASE
Vol. 154, No. 1
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Printed in U.S.A.

HALOPERIDOL, CLOPENTHIXOL, AND CHLORPROMAZINE IN CHRONIC SCHIZOPHRENIA

CHEMICALLY UNRELATED ANTIPSYCHOTICS AS THERAPEUTIC ALTERNATIVES

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In a double-blind placebo controlled clinical trial in chronic schizophrenic patients, haloperidol and clopenthixol were found to be effective antipsychotic agents, favorably altering behavior as observed by the psychiatrist, the nurse, and ward attendants. In this respect they appeared to compare adequately to the standard drug, chlorpromazine. In this experiment, neither the standard drug nor the investigational drugs affected psychological test performance.

The expected side effects such as extrapyramidal signs and sedation did occur and in some instances required dose reduction for alleviation. One case of possible hepatoxicity in the CX group occurred at the end of the study and was considered of serious import requiring cessation of medication.

It was concluded that these chemically unrelated antipsychotics could be used in practice as therapeutic alternatives, the particular choice depending on side effects and possibly symptomatology.

Since it was first made available for clinical use in 1960 (10), haloperidol (HAL), a butyrophenone, has been widely used as an antipsychotic drug, especially in Europe. In terms of chronic psychosis, HAL was thought to have "considerable clinical utility" after having been tried in this country as well (11). Canadian investigators (2) found it "at least as effective as the most potent phenothiazine drugs" in drug resistant psychotics.

Clopenthixol (CX), the thioxanthene analogue of perphenazine, was introduced as an antipsychotic drug in psychiatry during the last 10 years but its efficacy has been assessed in relatively few controlled studies in this country so far. Claims have been made regarding its usefulness in various categories of psychiatric patients, especially drug resistant schizophrenics with florid symptomatology, and also with respect to its low toxicity (1).

This report deals with the results of a double blind placebo controlled study of the therapeutic efficacy of HAL and CX in chronic schizophrenic patients using chlorpromazine (CPZ) as a standard reference drug. Of particular interest is the fact that three antipsychotic drugs representing three different chemical families are included in the same study and may be examined under the same experimental conditions.

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This research was supported in part by U.S. Public Health Service Grant MH 11666 and by National Institute of Mental Health Research Scientist Development Award K135278 (Dr. Serafetinides).

METHOD

Fifty-seven chronic schizophrenic subjects, 25 males and 32 females, ages 21 to 61, without evidence of organic disease, ill 2

TABLE 1
Dosage Information

	CX	CPZ	HAL	PL
Average daily dose overall	205 mg	830 mg	12.3 mg	8.7 capsules
Average dose last week of study	227 mg	945 mg	13.0 mg	10 capsules
Number on max. dose last week of study	12/15	10/11	12/14	13/13
Number receiving Cogentin*	8	4	6	3

* Benztropine mesylate, Merck Sharp & Dohme.

years or longer, were selected for study when permission from a responsible family member was obtained. The subjects were housed together on a special two-wing ward, males in one wing, females in the other, with a common day room between. The effects of previous treatment were allowed to dissipate over a 12-week[3] dry-out period during which time a physical examination was completed. Base line (pretreatment) measures were obtained during the 12th dry-out week and the subjects were randomly assigned to treatment. Each subject received medication from his own individual stock bottle, labeled only with his name. Starting with 2 capsules/day, the dose was increased by 1 capsule every 3 days to a maximum of 10 capsules, representing 15 mg of HAL, 250 mg of CX, 1000 mg of CPZ, or ten PL capsules. All medications were prepared in identically appearing capsules. Dose adjustment was permitted for the appearance of toxic symptoms or side effects and concomitant medication for Parkinsonism or bedtime sedation, when necessary, was also allowed. During the 12th week of medication, final (post-treatment) measures were obtained and the study was terminated.

MEASURES OF RESPONSE

1) Psychiatric assessments were based on individual interviews at the beginning and end of the treatment period permitting both clinical assessments of change (Clinical Global Impression (CGI) Severity of Illness and CGI Improvement) and the completion of the Overall and Gorham Brief Psychiatric Rating Scale (BPRS) (13). Background data were obtained by the project psychiatrist and the project social worker.

2) Ward behavior was observed daily by nursing service personnel (ward attendants) who completed the Nurse's Observation Scale for Inpatient Evaluation (NOSIE) (9) at 4-week intervals throughout the study and the Oklahoma Behavior Rating Scale (OBRS) (4) at the beginning and end of the treatment period. A global clinical impression of change was also supplied by the research nurse at the end of treatment.

3) A Psychological Test Battery was carried out at 4-week intervals throughout the study by a team of psychologists. It consisted of the following: Purdue Pegboard Digit Symbol, Stroop Color Word Test Muller-Lyer Illusion, Continuous Performance Test, Choice-Conflict Reaction Time Task, Finger Tapping Test, and Dotting Test.

4) Measured biological parameters included daily axillary body temperature; body weight and blood pressure measured once a week; urinalysis, hematocrit, white blood count (WBC) and differential, alkaline phosphatase, serum glutamic pyruvic aminase (SGPT), serum bilirubin, and blood urea nitrogen (BUN) measured every 4 weeks; serum cholesterol, serum triglycerides, and serum zinc measured at the beginning and end of treatment; slit lamp

[3] One subject (A.M.) because of rapid clinical deterioration with cessation of medication was removed from medication for only 2 weeks before the study.

TABLE 2
Demography

	CX (15)*	CPZ (14)	HAL (14)	PL (13)
Sex:				
Male	7	6	5	6
Female	8	8	9	7
Race:				
Caucasian	14	14	11	13
Other	1 Indian		1 Indian 2 Black	
Age:				
Average	40.9	40.1	43.4	43.9
Range	23–56	22–61	24–59	30–54
Married:				
Ever	2	6	7	5
Never	11	6	6	7
Not ascertained	1	2	1	1
Home situation:				
Conjugal	0	0	2	4
Nonconjugal	14	12	10	8
Not ascertained	1	2	2	1
Social Class:				
3	2	1	2	2
4	4	1	1	3
5	2	6	3	3
Not ascertained	7	6	8	5
History family mental illness	4	3	1	4
Previous hospitalization	13	10	11	11
Age at 1st hospitalization:				
<20	7	3	2	2
>20	8	11	12	11
Present hospitalization:				
<1 Year	1	1	1	2
>1 Year	14	13	13	11
Prestudy hospital adjustment:				
Vegetative and withdrawn	7	7	5	5
Some ward participation	7	4	5	7
Active ward participation	1	3	4	1
Duration of illness:				
Average	15	17	13	14
Range	6–34	5–30	3–30	6–33
Duration present episode illness:				
Average	14.81	13.1	9.05	13.26
Range	5.2–34	.6–29	1–28.7	4.5–22

* Number of patients in drug group.

examination carried out by consulting ophthalmologist at the beginning and end of treatment.

5) Toxicity and the occurrence of side effects was observed through regular ward rounds, and side effects were recorded on the Early Clinical Drug Evaluation Unit "Treatment Emergent Symptoms" (TES) form every 4 weeks.

6) The data were analyzed using Analysis of Variants (AOV) or Analysis of Covariance (AOCOV) for the overall F where initial and final mean scores were available. The Dunnett *t* test was then used for specific drug:PL comparisons. Significant regression effects precluding though invalidating the usual AOCOV mean comparisons, were considered effects in their own right. Significant mean comparisons in the face of heterogeneity of regression were validated by

AOV. The data derived from CGI were analyzed by χ^2 test.

RESULTS

Four of the 57 subjects, three on CPZ, and one on PL, failed to complete the 12 weeks of study. The PL subject and one CPZ subject were terminated because of behavioral deterioration after 4 and 8 weeks respectively. The other two CPZ subjects developed intestinal obstruction secondary to massive fecal impactions and were terminated in the 7th week of study.

The remaining 53 subjects completed the study and after 12 weeks, the average medication dose was HAL, 13 mg, CX, 227 mg, CPZ, 945 mg, and PL, 10 capsules. At that time 12 of 14 HAL subjects, 12 of 15 CX subjects, 10 of 11 CPZ subjects, and all of the 13 PL subjects were taking the maximum dose of 10 capsules per day (Table 1).

Demographic data for each treatment group are shown in Table 2. It appears that the treatment groups were relatively comparable for most aspects of their backgrounds.

Measures of response based on clinical global impression (CGI) were obtained from the psychiatrist and the research nurse. The CGI of the psychiatrist and the research nurse concerning overall improvement and the psychiatrist's CGI of Severity of Illness are shown in Tables 3 and 4, respectively.

No significant differences were found among treatments with either the psychiatric Severity of Illness rating or the psychiatric CGI of overall improvement, although a trend ($p < .10$) favoring drugs over PL was present. On the other hand, the CGI of improvement provided by the nurse, did differentiate both the investigational drugs and the standard drug from PL to a statistically significant degree. In this instance the

TABLE 3
CGI Improvement

	Psychiatrist Rated									
	Better			No change	Worse			$\bar{Y}$[a]	AOV	Per cent improved
	1	2	3	4	5	6	7			
CX (15)[b]	0	0	4	9	2	0	0	3.87	N.S.[c]	27
CPZ (11)	0	1	1	8	1	0	0	3.82		18
HAL (14)	0	0	4	6	4	0	0	4.00		29
PL (13)	0	0	0	7	6	0	0	4.46		0
	Nurse Rated									
	Better			No change	Worse			$\bar{Y}$	AOV	Per cent improved
	1	2	3	4	5	6	7			
CX (15)	1	3	7	4	0	0	0	2.93	$p < .01$	73
CPZ (12)	1	1	1	7	1	1	0	3.75		25
HAL (14)	2	1	3	6	1	1	0	3.43		43
PL (14)	0	0	0	7	3	3	1	4.86		0
	[d] (CX:PL, $p < .01$; CPZ:PL, $p < .025$; HAL:PL, $p < .01$)									

[a] $\bar{Y}$, final mean score.
[b] Number of patients in drug group.
[c] N.S., not significant.
[d] Dunnett t test.

TABLE 4
CGI Severity of Illness[a]

	Increasing Severity of Illness							$\bar{X}$[b]	$\bar{Y}$[c]	$\bar{Y}_a$[d]	AOCOV	Regression
	1	2	3	4	5	6	7					
CX:												
Initial	0	0	0	3	5	6	1	5.33				
Final	0	0	1	2	7	5	0		5.07	5.07		
CPZ:												
Initial	0	0	0	1	4	6	0	5.45				
Final	0	0	1	2	3	5	0		5.09	5.06	N.S.[e]	N.S.
HAL:												
Initial	0	0	1	1	6	5	1	5.29				
Final	0	0	0	2	7	5	0		5.21	5.22		
PL:												
Initial	0	0	0	3	4	6	0	5.23				
Final	0	0	0	1	2	10	0		5.69	5.71		

[a] Psychiatrist rated.
[b] $\bar{X}$, initial mean score.
[c] $\bar{Y}$, final mean score.
[d] $\bar{Y}_a$, adjusted final mean score.
[e] N.S., not significant.

HAL and the CX groups appeared to exhibit improvement in ward behavior, somewhat more frequently than did the CPZ group.

The mean, pre- and post-treatment, BPRS Total Scores for each treatment group are shown in Table 5. Significant Drug:PL differences were demonstrable for all three drug groups.

A breakdown of the BPRS by items is shown in Table 6, showing initial and final means, adjusted final means and significant Drug:PL comparisons by AOCOV. All three active drugs were significantly more effective than placebo for tension. CX and HAL were also more effective for anxiety and excitement, CX and CPZ for conceptual disorganization, and CPZ for hallucinatory behavior and unusual thought content.

The results obtained using the Nurse's Observation Scale for Inpatient Evaluation (NOSIE) are depicted in Table 7. The mean pre- and post-treatment scores and the adjusted post-treatment score for each treatment group for each factor and the total score are listed. Both investigational drug groups and the standard drug group showed significant Drug:PL differences on

TABLE 5
Brief Psychiatric Rating Scale (BPRS) Total Score

	$\bar{X}$[a]	$\bar{Y}$[b]	$\bar{Y}_a$[c]	AOCOV	Regression
CX	40.60	42.33	42.36		
CPZ	42.91	40.55	39.58		
HAL	38.64	41.64	42.51	$p < .01$	N.S.[d]
PL	41.00	52.08	51.93		
	[e](CX:PL, $p < .01$; CPZ:PL, $p < .01$; HAL:PL, $p < .01$)				

[a] $\bar{X}$, initial mean score.
[b] $\bar{Y}$, final mean score.
[c] $\bar{Y}_a$, adjusted final mean score.
[d] N.S., not significant.
[e] Dunnett t test.

Total Score and the Manifest Psychosis factor. In addition the CX group exhibited a significant regression effect as well as mean effect (AOV) on Irritability, suggesting that those subjects with greater initial irritability were most favorably affected by treatment with CX.

On the Oklahoma Behavior Rating Scale (Table 8), the subjects in both the HAL and CX drug groups exhibited significant Drug:PL mean effects on Total Extreme Behavior (E). In addition significant regression effects

TABLE 6

BPRS Items

	$\bar{X}$	$\bar{Y}$	$\bar{Y}_a$	AOCOV	Regression	Contrast
Somatic concern						
CX	1.00	1.67	1.77			
CPZ	1.09	1.09	1.11			
HAL	1.14	1.36	1.33			
PL	1.23	1.31	1.20	$p < .10$		
Anxiety						
CX	1.80	1.33	1.31			PL[d]
CPZ	1.82	2.09	2.06			
HAL	1.57	1.21	1.27			PL[d]
PL	1.77	2.62	2.61	$p < .01$		
Emotional withdrawal						
CX	2.73	3.07	3.01			
CPZ	2.82	3.18	3.09			
HAL	2.36	2.71	2.79			
PL	2.38	3.38	3.45			
Concept. disorganization						
CX	5.40	5.13	5.09			PL[d]
CPZ	5.45	5.18	5.14			PL[e]
HAL	5.00	5.36	5.39			
PL	4.92	6.00	6.05	$p < .05$		
Guilt						
CX	1.13	1.00	1.01			
CPZ	1.18	1.00	1.13			
HAL	1.29	1.00	1.00			
PL	1.77	1.08	1.05			
Tension						
CX	1.47	1.13	1.16			PL[d]
CPZ	1.55	1.36	1.36			PL[e]
HAL	1.50	1.14	1.16			PL[e]
PL	1.69	2.23	2.19	$p < .05$		
Mannerisms						
CX	1.33	1.60	1.66			
CPZ	1.91	1.55	1.49			
HAL	1.57	1.36	1.37			
PL	1.77	1.92	1.89			
Grandiosity						
CX	1.00	2.73	2.86			
CPZ	1.09	1.82	1.88			
HAL	1.00	2.36	2.49			
PL	1.62	2.31	1.95			
Depressed mood						
CX	1.47	1.20	1.21			
CPZ	1.18	1.00	.99			
HAL	1.21	1.00	1.00			
PL	1.31	1.00	1.00			
Hostility						
CX	1.93	1.80	1.80			
CPZ	2.18	2.00	2.02			
HAL	1.79	1.71	1.71			
PL	1.46	2.46	2.44			
Suspiciousness						
CX	2.07	1.80	1.77			
CPZ	2.45	2.00	1.87			
HAL	1.79	1.79	1.83			
PL	1.54	2.54	2.65			
Hallucinations						
CX	3.20	3.67	3.73			
CPZ	4.00	2.82	2.69			HAL,[e] PL[e]
HAL	2.86	4.42	4.56			
PL	3.92	4.77	4.66	$p < .05$		

TABLE 6—*continued*

	X	$\bar{Y}$	$\bar{Y}_a$	AOCOV	Regression	Contrast
Motor retardation						
CX	1.00	1.27	1.27			
CPZ	1.00	1.00	1.00			
HAL	1.00	1.00	1.00			
PL	1.00	1.00	1.00			
Uncooperative						
CX	1.67	1.67	1.68			
CPZ	1.36	1.18	1.18			
HAL	1.43	1.00	1.00			
PL	1.15	2.00	1.99			
Unusual thought						
CX	4.73	5.13	5.23			
CPZ	5.45	4.64	4.59			HAP,[e] PL[d]
HAL	5.36	5.43	5.40			
PL	5.39	5.77	5.73	$p < .05$		
Blunted affect						
CX	3.07	4.67	4.62			
CPZ	3.10	4.55	4.49			
HAL	2.86	4.71	4.71			
PL	2.46	4.92	5.03			
Excitement						
CX	1.33	1.00	1.06			PL[e]
CPZ	1.45	1.27	1.28			
HAL	1.50	1.14	1.14			PL[e]
PL	1.69	2.38	2.31	$p < .05$ (AOV)	$p < .05$	
Disorientation						
CX	4.27	2.47	2.29			
CPZ	3.82	2.82	2.83			
HAL	3.36	2.93	3.14			
PL	3.92	4.38	4.35			

[a] $\bar{X}$, initial mean score.
[b] $\bar{Y}$, final mean score.
[c] $\bar{Y}_a$, adjusted final mean score.
[d] $p < .01$.
[e] $p < .05$.

were demonstrable on Typical and Extreme Hyperactive Behavior (B+ and E+), most apparent for HAL on B+ and for CPZ on E+. In these instances those subjects exhibiting the most deviant behavior initially were the most affected by the drugs.

The entire psychological test battery including the Digit Symbol Test, Purdue Pegboard Test, Reaction Time Tests, the Dotting Test, Finger Tapping, Stroop Color-Word Test, Muller-Lyer Illusion Test, and Continuous Performance Tests, all failed to reflect significant treatment effects.

The Venables-O'Conner scale (16, 17) (Table 9) for rating withdrawal and paranoid schizophrenia reflected results as previously reported by Goldberg (7) in that the subjects on CPZ exhibited significant improvement on the withdrawal factor, but not the paranoid factor. The subjects on CX and HAL behaved similarly. However, a significant regression effect was demonstrable on the Paranoid Factor, most apparent for the HAL group, in which those subjects with high initial scores were most favorably affected by treatment.

Signs and symptoms of toxicity or undesirable pharmacological effects and abnormal laboratory values possible attributable to treatment are listed in Tables 10 and 11. It appears that extrapyramidal signs occurred somewhat more frequently with

TABLE 7

Nurse's Observation Scale for Inpatient Evaluation (NOSIE)

	$\bar{X}$[a]	$\bar{Y}$[b]	$\bar{Y}_a$[c]	AOCOV	Regression
Social competence					
CX	19.14	22.57	24.81		
CPZ	25.64	28.55	26.19	N.S.[d]	N.S.
HAL	22.57	25.86	25.78		
PL	22.62	18.92	18.70		
Social interest					
CX	15.38	16.15	16.37		
CPZ	16.55	18.73	18.01	N.S.	N.S.
HAL	18.14	20.14	18.14		
PL	12.71	17.71	20.07		
Neatness					
CX	15.00	18.29	18.80		
CPZ	15.09	17.82	18.27	N.S.	N.S.
HAL	17.86	21.43	20.14		
PL	15.08	14.46	14.92		
Irritability					
CX	25.29	18.14	15.08		
CPZ	18.91	15.27	16.22	N.S. (AOV,	$p < .05$
HAL	19.29	16.14	16.85	$p < .05$)	
PL	17.69	20.46	22.18		
	[a] CX:PL, $p < .01$				
Manifest psychosis					
CX	16.43	12.14	12.31		
CPZ	17.64	14.18	13.85	$p < .01$	N.S.
HAL	15.86	12.71	13.12		
PL	17.69	18.77	18.42		
	CX:PL, $p < .01$; CPZ:PL, $p < .01$; HAL:PL, $p < .01$				
Retardation					
CX	18.57	16.29	14.97		
CPZ	13.64	14.36	15.78	N.S.	N.S.
HAL	15.29	16.86	17.36		
PL	16.77	15.69	15.37		
Total score					
CX	106.43	134.14	143.43		
CPZ	127.09	141.27	136.90	$p < .01$	N.S.
HAL	127.43	143.86	138.80		
PL	120.69	114.69	114.23		
	CX:PL, $p < .01$; CPZ:PL, $p < .01$; HAL:PL, $p < .01$				

[a] $\bar{X}$, initial mean score.
[b] $\bar{Y}$, final mean score.
[c] $\bar{Y}_a$, adjusted final mean score.
[d] N.S., not significant.
[e] Dunnett t test.

HAL and CX than with CPZ. Transient sedation occurred often with CX and CPZ, but not at all with HAL. However, persistent sedation occurred in three instances with HAL and persisted as well in four subjects on CX, and one subject on CPZ. Orthostatic hypotension, significant constipation, and photosensitivity appeared only with CPZ. Significant weight loss occurred more frequently with HAL and PL (>10 lbs.) whereas significant weight gains (>10 lbs.) were more apparent with CPZ. Weight loss and weight gain occurred with equal frequency with CX.

Post-treatment eosinophilia seemed to occur slightly more often with HAL and CX

TABLE 8

Oklahoma Behavior Rating Scale (OBRS)

	$\bar{X}^a$	$\bar{Y}^b$	$\bar{Y}_a{}^c$	AOCOV	Regression
"B+"					
CX	7.0	1.7	1.13		
CPZ	6.5	1.6	1.26	N.S.[d] (AOV)	$p < .01$
HAL	2.4	0.6	0.79		
PL	6.2	4.7	6.40		
"E"					
CX	43.1	31.6	26.56		
CPZ	36.4	29.7	29.63		
HAL	31.7	26.4	29.74	$p < .05$	N.S.
PL	33.6	36.2	38.14		
		[e] CX:PL $p < .01$; HAL:PL $p < .05$			
"E+"					
CX	12.7	4.4	3.82		
CPZ	13.9	4.5	3.50	N.S. (AOV)	$p < .01$
HAL	6.6	3.2	4.74		
PL	11.8	12.0	11.75		

[a] $\bar{X}$, initial mean score.
[b] $\bar{Y}$, final mean score.
[c] $\bar{Y}_a$, adjusted final mean score.
[d] N.S., not significant.
[e] Dunnett t test.

TABLE 9

Venables-O'Conner Scale

	$\bar{X}^a$	$\bar{Y}^b$	$\bar{Y}_a{}^c$	AOCOV	Regression
Paranoid factor					
CX	9.00	6.40	6.16		
CPZ	7.82	7.09	7.20	N.S.[d] (AOV)	$p < .05$
HAL	7.36	6.79	7.04		
PL	8.46	8.08	8.00		
Withdrawal factor					
CX	23.80	20.87	20.48		
CPZ	23.64	20.00	19.72	$p < .05$	N.S.
HAL	22.14	19.86	20.57		
PL	23.31	24.15	24.08		
	[e] CX:PL, $p < .01$; CPZ:PL, $p < .01$; HAL:PL, $p < .05$				

[a] $\bar{X}$, initial mean score.
[b] $\bar{Y}$, final mean score.
[c] $\bar{Y}_a$, adjusted final mean score.
[d] N.S., not significant.
[e] Dunnett t test.

than with CPZ. Low serum cholesterol concentrations seemed to occur more often with HAL, whereas elevated triglycerides seemed more often to be associated with CX. SGPT alterations occurred only with CX in this study, and in one instance this was persistent and progressive and resulted in withholding medication and a complete workup for liver disease including a needle biopsy. The biopsy was interpreted as steatosis and possible early cirrhosis, but its relationship to drug treatment is difficult to assess. The patient subsequently made a complete clinical recovery, but unfortunately a follow-up liver biopsy could not be obtained. Otherwise none of the laboratory

changes was considered of sufficient importance to alter treatment.

DISCUSSION

Although minor differences seem apparent among the three drug treatments, their similarity in terms of overall antipsychotic efficacy is perhaps the most striking finding in this study. Indeed, trying to differentiate treatments on the basis of antipsychotic activity alone would be most difficult. This is remarkable in view of the fact that these drugs are not chemically related. In addition to the theoretical issues involved, the practical implications of the finding are that these drugs can be used as alternatives in the treatment of chronic schizophrenic patients. Thus, the particular choice will be made largely on the basis of the side effects of the drug in question.

The demographic data indicate that the average subject in this study might be described as a 40-year-old Caucasian male or female, probably never married, who had been ill for more than 15 years. Prior to study the subject was on the vegetative or withdrawn side, participating in only a limited fashion in ward activities. The subject's family and home situation were frequently unfavorable and the present episode of illness was generally continuous

TABLE 10

Reported Side Effects Emerging with Treatment, (Exclusive of Clinical Laboratory Tests)

	CX	CPZ	HAL	PL
Extrapyramidal signs..	7 (1 akathisia)	3	6	0
Excitement	1	0	0	0
Sedation				
Transient	6	7	0	2
Persistent	4	1	3	0
Orthostatic symptoms				
Transient	0	4	0	0
Persistent	0	1	0	0
Constipation	0	2*	0	0
Weight:				
10 lb. loss	2	2	6	7
10 lb. gain	2	4	0	0
Photosensitivity	0	2	0	0
Ankle edema	0	0	0	1

* Severe, simulating obstruction—surgical exploration in one case showed impaction only.

TABLE 11

Abnormal Laboratory Values Emerging with Treatment

	CX	CPZ	HAL	PL
Urine:				
>10 WBC/hpf	2	0	1	1
Sugar	0	1 (Trace)	0	0
Hematology				
WBC > 12000	1	0	0	1
Eos. > 5 per cent	3	1	3	1
WBC < 4000	0	1	0	0
Liver				
SGPT > 49 Units	3*	0	0	0
Total bilirubin > 1.4 mg per cent	1	0	0	0
BUN				
>20 mg per cent	0	0	1	0
Cholesterol				
>300 mg per cent	0	0	0	0
<170 mg per cent	1 (8)†	1 (2)	3 (6)	0 (6)
Triglycerides				
>4.6 mEq./l	3 (7)	1 (3)	1 (5)	1 (5)
Zinc				
<70 μg per cent	0	0	0	1
>120 μg per cent	1	0	1	0

* One case, steatosis and ? cirrhosis on needle biopsy of liver. Clinical recovery with medication withdrawal—no biopsy follow-up. Two cases, transient elevation; asymptomatic; medication not stopped.

† Numbers in parentheses indicate incidence of abnormal values at base line.

over more than 12 years. Thereby, the marked chronicity of illness of these subjects was established and the comparability of the four treatment groups on these variables was relatively good.

It is of interest that the total BPRS score reflected Drug:PL differences for CX, HAL, and CPZ, but this appeared to be principally a function of the deterioration of the PL group. Only the CPZ group scores indicated slight overall improvement. However, a breakdown of the BPRS by items did bring out drug-induced improvement in some areas with HAL and CX appearing more effective for a constellation of secondary symptoms such as tension, anxiety, and excitement while CPZ appeared more effective for a constellation of more primary symptoms such as conceptual disorganization and unusual thought content. Interestingly, this is at variance with the findings of Gonnier, Schiele, and Vestre (8) who reported recently that HAL was more effective in reducing typically psychotic symptoms such as conceptual disorganization. Although, these investigators also used the BPRS in their assessment, comparison is difficult, since their control drug was thioridazine and their study populations consisted of male patients only.

The NOSIE and OBRS which reflect ward behavior as seen by the ward attendants in this study, rather than the research nurse or psychiatrist, did depict significant Drug: PL differences on Manifest Psychosis, Irritability, and Total Score of the NOSIE and B+, E and E+ of the OBRS, based on positive improvement trends for the drug groups rather than negative trends for the PL group.

The failure of the psychological test battery to reflect drug effect even in part is most striking. The fact that the numbers of groups were reduced by those who failed to participate either at the beginning or the end may account for this. The observation scales, on the other hand, do not require patient cooperation or participation.

On the Venables-O'Conner scale (16, 17), CX and HAL as well as CPZ showed the expected improvement in withdrawal, and CPZ as expected showed no mean change on the Paranoid factor. This was in agreement with the observations of Goldberg (7). However, the significant regression effect on the Paranoid factor, suggesting differential treatment effects for high and low initial scores, appears to involve all treatments, including the CPZ and thereby clouds the issue.

The incidence of extrapyramidal signs was higher with HAL and CX than with CPZ. This higher incidence among the HAL patients in our study is in keeping with the findings of other investigators (6). However, none of these signs were of a particularly severe or dystonic type. Claims have been made to the effect that whereas extrapyramidal signs occur with greater frequency at higher doses (above 5 mg, daily), the best therapeutic effects are seen with relatively low doses (up to 5 mg, daily) (11). However other reports stress that higher doses are necessary for good therapeutic results (6).

Excessive sedation, as a transient side effect, was quite as noticeable with CX in this study as with CPZ. Simpson *et al.* (15) reported sedation with low doses CX, presumably early; Merlis (12) also reported "initial severe sedation" in all his 12 subjects who reached a maximum dose of 150 mg daily. In the present study, excessive sedation after an initial dose of 50 mg CX was noted in 10 of 15 subjects. In all but four instances this was a transient phenomenon disappearing over a period of a few days to a few weeks. In those four instances attempts to adhere to the dose increment prescribed by the study protocol were unsuccessful and eventually adjustments had to be made to control the sedation. Three instances of persistent sedation also occurred with HAL requiring dosage adjustment; with CPZ, sedation persisted in only one case. On the other hand, the restlessness and insomnia reported by Schiele (14) and the photophobia

and high incidence of akathisia mentioned by Merlis (12) were not observed with CX. However, a relatively high incidence of extrapyramidal signs other than akathisia was noted.

Finally, our tentative low serum cholesterol values for HAL, although needing further investigation, are noted with interest in view of the hypocholesterolemic effects of another butyrophenone [trifluperidol (3)] and the hypercholesterolemic effects of the phenothiazine chlorpromazine (5).

In conclusion, it can probably be said that due to their overall antipsychotic similarity, these three drugs may be considered as therapeutic alternatives and prescribed as such despite their chemical dissimilarity. The particular choice would depend partly on the side effects in question and possibly also on the type of symptomatology present.

REFERENCES

1. Ayd, F. J. Current status of clopenthixol. Int. Drug Ther. Newsletter *4:* 5–8, 1969.
2. Ban, T. A. and Lehmann, H. E. Efficacy of haloperidol in drug refractory patients. Int. J. Neuropsychiat. *3:* 78–83, 1967.
3. Clark, M. L., Braun, G. A., Hewson, J. R., Serafetinides, E. A., Colmore, J. P. and Rahhal, D. K. Trifluperidol and cholesterol in man. Clin. Pharmacol. Ther. *9:* 333–340, 1968.
4. Clark, M. L., Ray, T. S., Paredes, A., Costiloe, J. P., Chappell, J. S., Hagans, J. A. and Wolf, S. Chlorpromazine in chronic schizophrenic women: I. Experimental design and effects at maximum point of treatment. Psychopharmacologia *2:* 107–136, 1961.
5. Clark, M. L., Ray, T. S., Paredes, A., Ragland, R. E., Costiloe, J. P., Smith, C. W. and Wolf, S. Chlorpromazine in women with chronic schizophrenia. The effect on cholesterol levels and cholesterol-behavior relationships. Psychosom. Med. *29:* 634–642, 1967.
6. Crane, G. E. A review of clinical literature on haloperidol. Int. J. Neuropsychiat. *3:* 110–123, 1967.
7. Goldberg, S. C. Prediction of response to antipsychotic drugs. Presented at the American College of Neuropsychopharmacology. San Juan, Puerto Rico, December, 1967.
8. Gonier, T., Schiele, B. C. and Vestre, N. D. A comparison of haloperidol and thioridazine HCl in chronic treatment-resistant schizophrenics. Physicians' Drug Manual *2:* 47–54, 1970
9. Honigfeld, G. and Klett, C. The nurse's observation scale for inpatients evaluation (NOSIE). A new scale for measuring improvement in chronic schizophrenia. J. Clin. Psychol. *21:* 65–71, 1965.
10. Janssen, P. A. J. The pharmacology of haloperidol. Int. J. Neuropsychiat. *3:* 10–18, 1967.
11. Lapolla, A. Clinical efficacy of haloperidol in chronic psychiatric patients. Int. J. Neuropsychiat. *3:* 68–74, 1967.
12. Merlis, S. Thioxanthenes: Discussion of the clinical investigation papers. In *The Thioxanthenes*, pp. 113–116. Karger, New York, 1969.
13. Overall, J. E. and Gorham, D. R. Brief psychiatric rating scale. Psychol. Rep. *10:* 799–822, 1962.
14. Schiele, B. C. Studies with clopenthixol and thiothixene. In *The Thioxanthenes*, pp. 100–105. Karger, New York, 1969.
15. Simpson, G. M., Angus, J. W. S. and Igbal, I. The effect of clopenthixol in chronic schizophrenia. Curr. Ther. Res. *8:* 85, 1966.
16. Venables, P. H.: A short scale for rating "activity-withdrawal" in schizophrenics. J. Ment. Sci. *103:* 197–199, 1957.
17. Venables, P. H. and O'Connor, N. A short scale for rating paranoid schizophrenia. J. Ment. Sci. *105:* 815–818, 1959.